Dynamic Emergency Response Allocation Agent

CS 498 AI Agents in the Wild - Final Presentation

A hybrid agent that combines lookahead planning with conditional LLM arbitration for zero-error emergency dispatch.

The Problem: LLMs Can't Optimize Under Hard Constraints

LLMs Excel At

Semantic reasoning, parsing unstructured incident reports, and generating natural language justifications.

LLMs Fail At

Strict mathematical optimization — they hallucinate routes, lose track of capacities, and output physically infeasible actions.

Disaster Response Requires Both

Zero-error spatial constraint satisfaction *and* unstructured text interpretation — simultaneously, under deadline.

End-to-end LLM solvers fail catastrophically on hard constraints. Our approach: **LLM as orchestrator, not calculator.**

RescueBench: A Deterministic Emergency Dispatch Benchmark

What We Built

- 20 deterministic JSON scenarios on a shared 16-node city map
- Heterogeneous 4-class fleet: ambulances, fire engines, police, supply trucks
- Deadlines verified via Dijkstra's algorithm: no subjective grading

How We Generated It

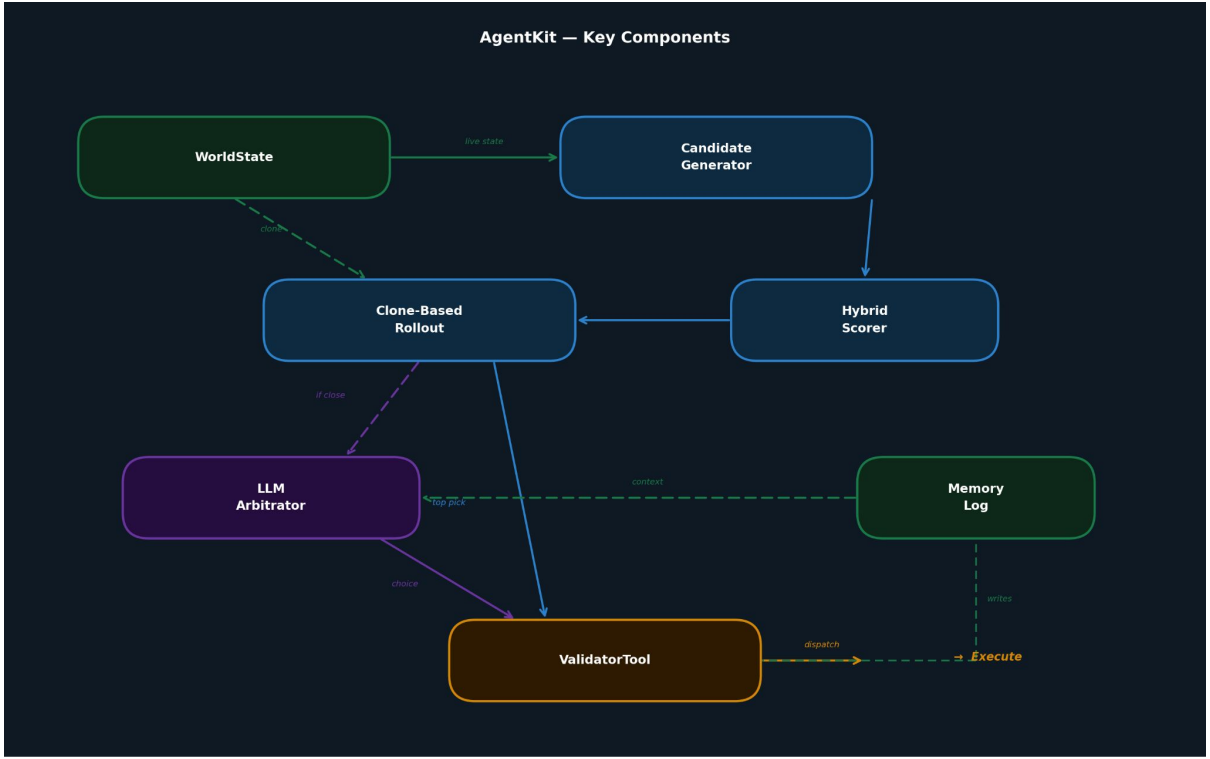
- Manually authored base city graph guaranteeing solvability
- LLM-assisted mutation pipeline (Gemini) for diverse incidents
- Injected timestamped infrastructure failures: bridge collapses, road floods

Four Tiers, Four Failure Modes

Tier	Challenge	What It Tests
1 — Basic Triage	Direct routing, single-unit dispatches	Sanity check
2 — Constraint Math	Capacity arithmetic, multi-unit coordination	Feasibility reasoning
3 — Ethical Prioritization	Forced scarcity — abandon low-severity to save high-severity	Global optimization
4 — Dynamic Replanning	Mid-mission infrastructure failures triggered by simulation clock	Adaptive replanning

Tier 3 example: Train derailment needs all 4 medical units. Two minor injuries also active. Agent must *intentionally* abandon them to maximize global score. **Tier 4 example:** At t=8 min, suspension bridge blocked — agent must recalculate all routes on the fly.

AgentKit Architecture:

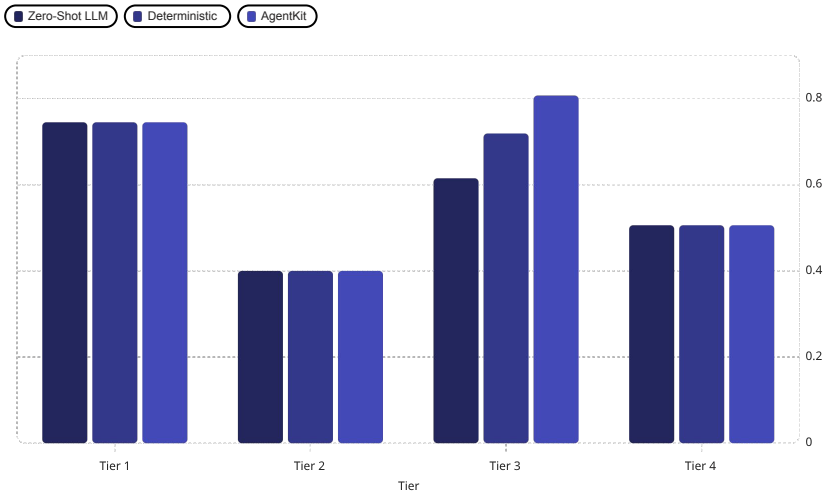


After each dispatch, the agent waits for the next event — a vehicle return or a dynamic trigger. On alert, it flushes its queue and re-ranks all open incidents from scratch. The LLM is invoked **only** when top candidate scores are within 8% of each other.

Seven Components, One Hybrid Agent

1	**WorldState** Live simulation — vehicles, incidents, routes, hospitals
2	**Candidate Generator** Enumerates every valid (vehicle → incident) pair
3	**Hybrid Scorer** Ranks candidates by urgency, slack, and scarcity
4	**Clone-Based Rollout** Deep-copies simulator, projects future PWRS per candidate
5	**ValidatorTool** Hard gate — nothing dispatches without passing constraints
6	**LLM Arbitrator** Tiebreaker invoked only when top scores are within 8%
7	**Memory Log** Feeds recent decisions and alerts into LLM context window

Results: AgentKit Never Regresses



Priority-Weighted Resolution Score (PWRS)

AgentKit achieves the highest mean PWRS (0.615) and is the **only method that never regresses** across any tier.

Tier 3: +8.8pp over deterministic, +19.2pp over zero-shot

- Zero-shot introduces constraint violations on Tier 2
- Tier 4: all methods fail equally — an environmental ceiling

⚠ Zero-shot Tier 2 PWRS = Cap-PWRS (0.400 = 0.400): zero partial credit, binary outcomes only — structural failure.

Where AgentKit Wins: Sequencing and Coordination

Tier 3, Scenario 13 — Vehicle Sequencing

One police unit, two incidents: a riot (18.3 min away) and a traffic call (6.7 min away).

Method	Order	PWRS
Zero-Shot	Riot first	0.850
Deterministic	Riot first	0.850
AgentKit	**Traffic → Riot**	**1.000**

Rollout projected that the fast job first returns the unit in time for the riot deadline.

Tier 3, Scenario 15 — Trolley Problem

Three trolley incidents, one police unit, one medic. Zero-shot cannot coordinate parallel dispatch without planning structure.

Method	PWRS
Zero-Shot	0.375
Deterministic	1.000
AgentKit	**1.000**

Where Everyone Struggles: The Infrastructure Ceiling

Tier 4, Scenario 16 — Bridge Collapse Mid-Mission

MED_01 dispatched at t=0. Bridge collapses at t=8, forcing a major detour. Arrival at t=45.3 — far past deadline. **All methods: PWRS = 0.000.** Replanning only affects *future* dispatches: committed vehicles cannot be recalled.

Tier 4, Scenario 19 — Simultaneous Double Blockage

POLICE_01 en route to riot. At t=10: northern approach iced, southern barricaded. After completing the mission, no valid path remains to any open incident. **All methods: PWRS = 0.000.**

Lookahead only helps when *sequencing* matters — Tier 3 gains, not Tier 4. The hard ceiling is infrastructure, not algorithm.

Conclusion: Structure Beats Raw LLM

1. **Pre-validated candidates eliminate violations**

   Zero-shot fails structurally on hard constraints: deterministic grounding is non-negotiable.

2. **Lookahead is the real performance driver**

   Tier 3 gains came from clone-based rollout, not LLM arbitration: the LLM confirmed the heuristic every time.

3. **Tier 4 is an infrastructure ceiling, not a planning problem**

   When the environment is adversarially constrained, no planning overhead helps.

Future work: Bundle-level joint planning, predictive replanning before dynamic events fire, and stress-testing LLM value in genuinely ambiguous, information-sparse scenarios.